EXP1

Exploring Matplotlib, Seaborn and Plotly Visualization Tools

AIM:

To explore the Matplotlib, Seaborn and Plotly visualization libraries by loading the Titanic dataset, inspecting and cleaning it, and drawing a bar chart, pie chart, histogram, box plot and scatter plot.

ALGORITHM:

1. Import the required libraries: pandas, numpy, matplotlib and seaborn.
2. Load the Titanic dataset (Titanic-Dataset.csv) into a DataFrame using read_csv().
3. Display the first five and last five rows using head() and tail().
4. Check the size of the data using shape, and its structure using info() and describe().
5. Find the number of missing values in each column using isnull().sum().
6. Fill the missing Age values with the median, fill the missing Embarked values with the mode, and drop the Cabin column.
7. Check the missing values again to confirm that the data is clean.
8. Draw a bar chart of the number of passengers in each class using Plotly Express.
9. Draw a pie chart of survived and not survived passengers using the pandas plot (Matplotlib) function.
10. Draw a histogram with a density curve of passenger age using Seaborn.
11. Draw a box plot of passenger age using Seaborn.
12. Draw a scatter plot of Age against Fare, coloured by Survived, using Seaborn.
13. Observe the charts and note the results.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("Titanic-Dataset.csv")
```

```
print(df.head())
```

```
print(df.tail())
```

```
print("Shape:", df.shape)
```

```
df.info()
```

```
df.describe()
```

```
print(df.isnull().sum())
```

```
df["Age"] = df["Age"].fillna(df["Age"].median())

# Fill missing Embarked values with the mode
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

# Drop Cabin column
df.drop(columns=["Cabin"], inplace=True)
```

```
print(df.isnull().sum())
```

```
import plotly.express as px

fig = px.bar(
    x=df["Pclass"].value_counts().sort_index().index,
    y=df["Pclass"].value_counts().sort_index().values,
    labels={
        "x": "Passenger Class",
        "y": "Number of Passengers"
    },
    title="Number of Passengers by Class"
)

fig.show()
```

```
df["Survived"].value_counts().plot(
    kind="pie",
    figsize=(6,6),
    labels=["Not Survived", "Survived"],
    autopct="%1.1f%%",
    startangle=90,
    shadow=True
)

plt.title("Passenger Survival Distribution")
plt.ylabel("")

plt.show()
```

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))

sns.histplot(
    data=df,
    x="Age",
    bins=20,
    kde=True
)

plt.title("Distribution of Passenger Age")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt

plt.figure(figsize=(6,5))

sns.boxplot(data=df, y="Age")

plt.title("Box Plot of Passenger Age")
plt.ylabel("Age")

plt.show()
```

```
plt.figure(figsize=(8,5))

sns.scatterplot(
    data=df,
    x="Age",
    y="Fare",
    hue="Survived"
)

plt.title("Age vs Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")

plt.show()
```

OUTPUT:

Output of: print(df.head())

```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

Output of: print(df.tail())

```
     PassengerId  Survived  Pclass                                      Name  \
886          887         0       2                     Montvila, Rev. Juozas
887          888         1       1              Graham, Miss. Margaret Edith
888          889         0       3  Johnston, Miss. Catherine Helen "Carrie"
```

```
889          890         1       1                      Behr, Mr. Karl Howell
890          891         0       3                        Dooley, Mr. Patrick

        Sex   Age  SibSp  Parch      Ticket   Fare Cabin Embarked
886    male  27.0      0      0      211536  13.00   NaN        S
887  female  19.0      0      0      112053  30.00   B42        S
888  female   NaN      1      2  W./C. 6607  23.45   NaN        S
889    male  26.0      0      0      111369  30.00  C148        C
890    male  32.0      0      0      370376   7.75   NaN        Q
```

Output of: print("Shape:", df.shape)

```
Shape: (891, 12)
```

Output of: df.info()

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   PassengerId   891 non-null    int64
 1   Survived      891 non-null    int64
 2   Pclass        891 non-null    int64
 3   Name          891 non-null    str
 4   Sex           891 non-null    str
 5   Age           714 non-null    float64
 6   SibSp         891 non-null    int64
 7   Parch         891 non-null    int64
 8   Ticket        891 non-null    str
 9   Fare          891 non-null    float64
 10  Cabin         204 non-null    str
 11  Embarked      889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 118.9 KB
```

Output of: df.describe()

```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

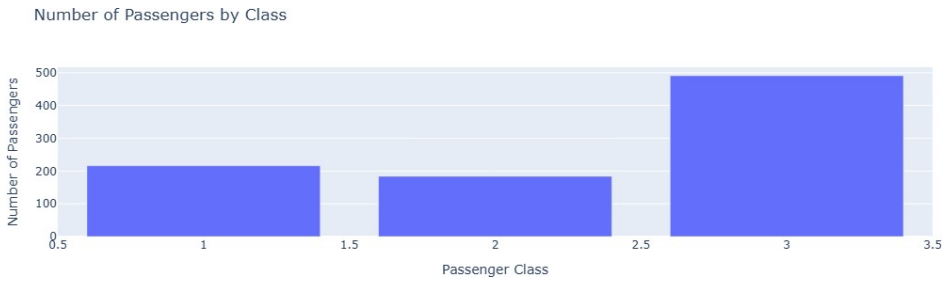
Output of: print(df.isnull().sum())

```
PassengerId       0
Survived          0
Pclass            0
Name              0
Sex               0
Age             177
SibSp             0
Parch             0
Ticket            0
Fare              0
Cabin           687
Embarked          2
dtype: int64
```

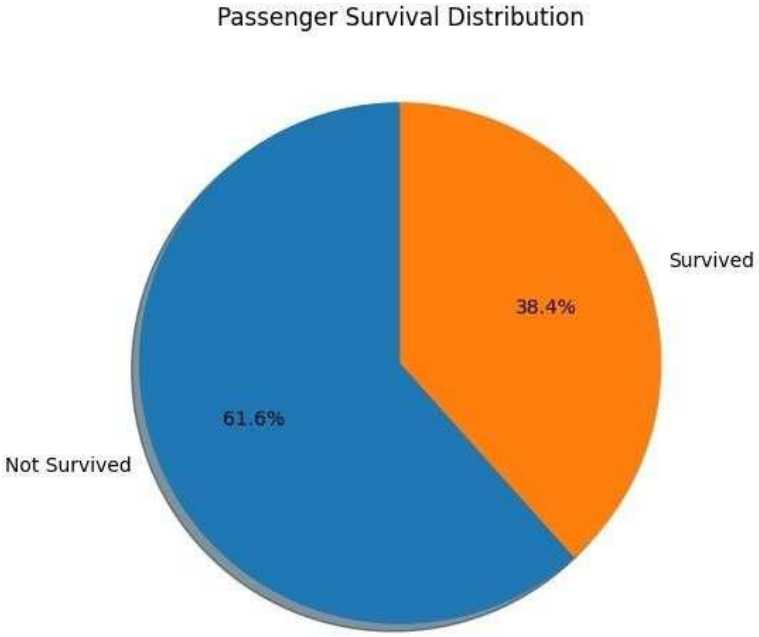
Output of: print(df.isnull().sum())

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

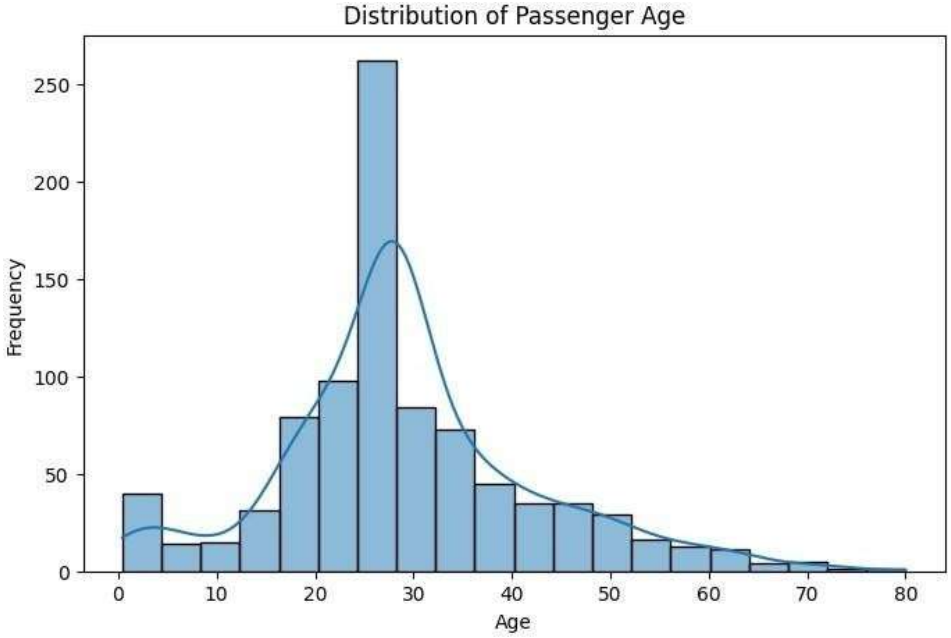
Output of: import plotly.express as px



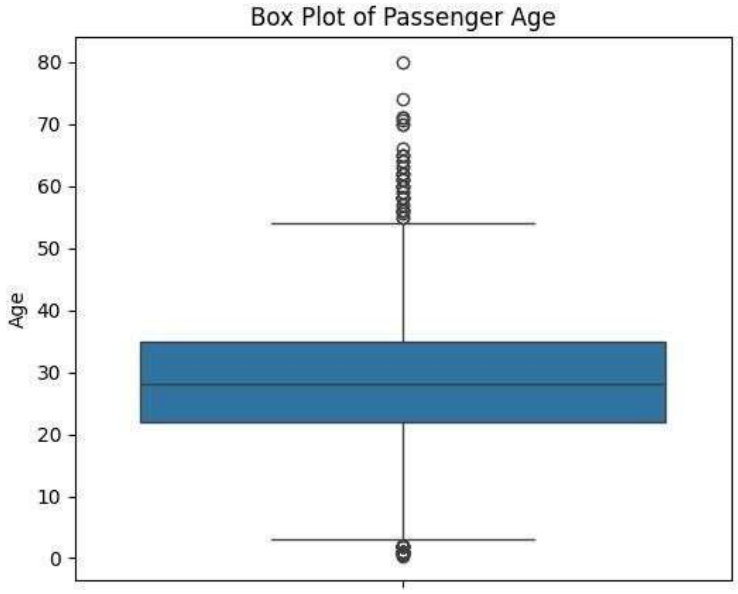
Output of: df["Survived"].value_counts().plot(



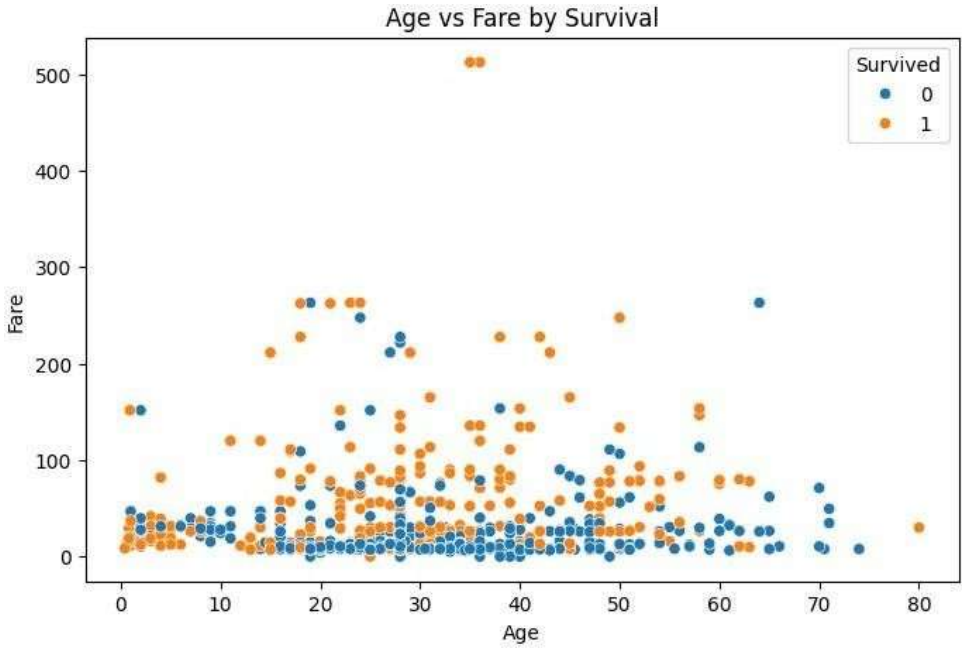
Output of: import seaborn as sns



Output of: import seaborn as sns



Output of: plt.figure(figsize=(8,5))



RESULT:

Thus, the Matplotlib, Seaborn and Plotly libraries were explored using the Titanic dataset (891 rows and 12 columns). The missing values in Age (177), Embarked (2) and Cabin (687) were handled, and the bar chart, pie chart, histogram, box plot and scatter plot were drawn successfully.